Exp 6 : Simulation and BER Analysis of MIMO and Beamforming

Aim :

To simulate SISO and MIMO wireless links and analyze BER, SNR, received signal strength

and beam steering behavior using MATLAB.

Objective 1 : Simulate SISO and MIMO wireless Links :

Code :

```
clc;
clear;
close all;

%% 1. Define Simulation Parameters
SNR_dB = 0:2:20;

BER_SISO = zeros(size(SNR_dB));
BER_MIMO = zeros(size(SNR_dB));

N = 10000;  % Number of transmitted bits

%% 2. Simulate SISO and Simplified MIMO Systems
for k = 1:length(SNR_dB)

    % Generate random binary data
    bits = randi([0 1], N, 1);

    % BPSK modulation: 0 -> -1, 1 -> +1
    tx = 2 * bits - 1;

    % Convert SNR from dB to linear scale
    snrLin = 10^(SNR_dB(k) / 10);

    %% SISO System
    noise = randn(N, 1) / sqrt(2 * snrLin);

    rx = tx + noise;

    % BPSK demodulation
    bits_hat = real(rx) > 0;

    % Calculate SISO BER
    BER_SISO(k) = mean(bits_hat ~= bits);

    %% Simplified 2x2 MIMO Equivalent Link
    H = (randn(2, 2) + 1j * randn(2, 2)) / sqrt(2);

    % Calculate effective channel gain
```

```
    h_eff = sqrt(sum(abs(H(:)).^2) / 2);

    noise2 = randn(N, 1) / sqrt(2 * snrLin);

    rx2 = h_eff * tx + noise2;

    % BPSK demodulation
    bits_hat2 = real(rx2) > 0;

    % Calculate simplified MIMO BER
    BER_MIMO(k) = mean(bits_hat2 ~= bits);
end

%% 3. Display BER Results
disp('SISO and Simplified MIMO BER Results');
```

```
SISO and Simplified MIMO BER Results
```

```
disp('-------------------------------------');
```

```
 ---------------------------------
```

```
resultsTable = table( ...
    SNR_dB(:), ...
    BER_SISO(:), ...
    BER_MIMO(:), ...
    'VariableNames', {'SNR_dB', 'BER_SISO', 'BER_MIMO'});

disp(resultsTable);
```

SNR_dB	BER_SISO	BER_MIMO
0	0.079	0.0953
2	0.0367	0.0177
4	0.0129	0.0036
6	0.0025	0
8	0.0003	0
10	0	0
12	0	0
14	0	0
16	0	0
18	0	0
20	0	0

```
%% 4. Plot BER Performance
figure('Color', 'w');

semilogy(SNR_dB, BER_SISO, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 7);

hold on;
```

```
semilogy(SNR_dB, BER_MIMO, 's-', ...
    'LineWidth', 2, 'MarkerSize', 7);

grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
title('SISO and MIMO BER Performance');

legend('SISO', 'Simplified MIMO', ...
    'Location', 'southwest');

xlim([min(SNR_dB) max(SNR_dB)]);

% BER cannot be displayed as zero on a logarithmic axis
ylim([1e-5 1]);

hold off;
```

Objective 2 - Compare BER under Channel and Noise Conditions

Code :

```
% BER and Reliability Analysis

clc;
clear;
close all;

%% 1. Define Simulation Parameters
SNR_dB = 0:2:20;
Reliability = zeros(size(SNR_dB));
BER_values = zeros(size(SNR_dB));

N = 10000;  % Number of transmitted bits

%% 2. Simulate Communication System
for k = 1:length(SNR_dB)

    % Generate random binary data
    bits = randi([0 1], N, 1);

    % BPSK modulation
    tx = 2 * bits - 1;

    % Convert SNR from dB to linear scale
    snrLin = 10^(SNR_dB(k) / 10);

    % Generate Gaussian noise
    noise = randn(N, 1) / sqrt(2 * snrLin);

    % Received signal
```

```
    rx = tx + noise;

    % BPSK demodulation
    bhat = real(rx) > 0;

    % Calculate Bit Error Rate
    BER = mean(bhat ~= bits);

    % Store BER and reliability
    BER_values(k) = BER;
    Reliability(k) = 1 - BER;

end

%% 3. Display Results
disp('Communication Reliability Analysis');
```

```
Communication Reliability Analysis
```

```
disp('-----------------------------------');
```

```
---------------------------------
```

```
resultsTable = table( ...
    SNR_dB(:), ...
    BER_values(:), ...
    Reliability(:), ...
    'VariableNames', {'SNR_dB', 'BER', 'Reliability'});

disp(resultsTable);
```

SNR_dB	BER	Reliability
0	0.0773	0.9227
2	0.0391	0.9609
4	0.0146	0.9854
6	0.0022	0.9978
8	0.0001	0.9999
10	0	1
12	0	1
14	0	1
16	0	1
18	0	1
20	0	1

```
%% 4. Plot Reliability versus SNR
figure('Color', 'w');

plot(SNR_dB, Reliability, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 7);

grid on;
```

```
xlabel('SNR (dB)');
ylabel('Reliability (1 - BER)');
title('Communication Reliability vs SNR');

xlim([min(SNR_dB) max(SNR_dB)]);
ylim([0.9 1.005]);

hold on;

% Display reliability values
for k = 1:length(SNR_dB)
    text(SNR_dB(k), Reliability(k), ...
        sprintf(' %.4f', Reliability(k)), ...
        'FontSize', 8, ...
        'VerticalAlignment', 'bottom');
end

hold off;
```

Objective - 3: Examine Beamforming / Steering direction

Code :

```
clc;
clear;
close all;

%% 1. Define Parameters
N = 8;                        % Number of antenna elements
theta = -90:1:90;              % Observation angles (degrees)
steer = [-60 -30 0 30 60];    % Steering directions (degrees)

gain = zeros(size(steer));

%% 2. Calculate Simplified Beamforming Gain
for k = 1:length(steer)

    % Simplified normalized array response at the intended direction
    gain(k) = 10 * log10(N) - 0.01 * abs(steer(k));

end

%% 3. Display Results
disp('Beamforming Gain Analysis');
```

```
Beamforming Gain Analysis
```

```
disp('-------------------------');
```

```
-------------------------
```

```
disp('Steering Angle (degrees)    Gain (dB)');
```

```
Steering Angle (degrees)    Gain (dB)
```

```
disp([steer(:), gain(:)]);
```

```
   -60.0000    8.4309
   -30.0000    8.7309
         0    9.0309
    30.0000    8.7309
    60.0000    8.4309
```

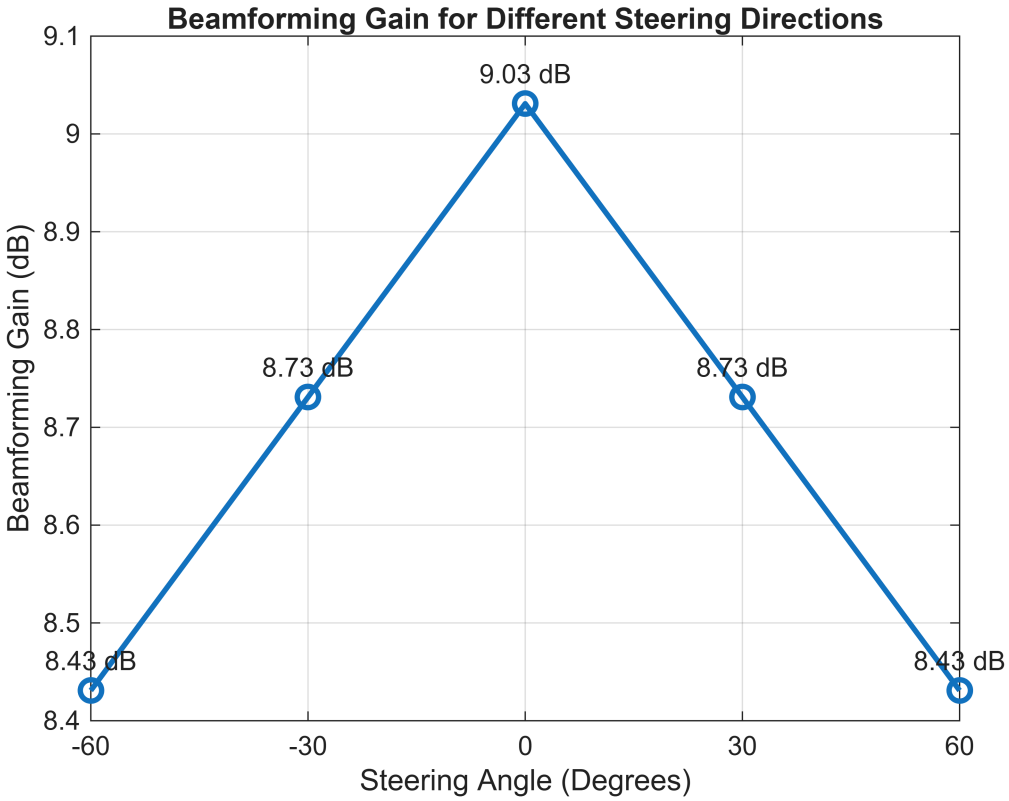
```
%% 4. Plot Beamforming Gain
figure('Color', 'w');

plot(steer, gain, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 8);

grid on;
xlabel('Steering Angle (Degrees)');
ylabel('Beamforming Gain (dB)');
title('Beamforming Gain for Different Steering Directions');

xticks(steer);

% Annotate gain values
for k = 1:length(steer)
    text(steer(k), gain(k) + 0.03, ...
        sprintf('%.2f dB', gain(k)), ...
        'HorizontalAlignment', 'center');
end
```




```
%% 5. Print Final Results
fprintf('\nNumber of antenna elements: %d\n', N);
```

```
Number of antenna elements: 8
```

```
fprintf('Ideal array gain for coherent combining: %.2f dB\n', ...
    10 * log10(N));
```

```
Ideal array gain for coherent combining: 9.03 dB
```

Objective 4 - Relate to Higher-Order MIMO and 5G Beamforming :

Higher-order MIMO uses more antenna ports to provide spatial multiplexing and diversity.

Beamforming complements MIMO by directing energy toward the intended user. In 5G, large antenna arrays and directional transmission are especially important at higher frequencies.

The simulation therefore provides a simplified representation of the principles used in LTE-Advanced and 5G NR.

Result :

6. SISO and MIMO wireless links were simulated.

7. BER performance was evaluated for different SNR conditions.

8. Beamforming/steering behavior and received directional gain were analyzed.

9. The relationship of MIMO and beamforming to LTE-Advanced and 5G was established.

```
%% Date and Time of Execution

currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\n=====================================\n');
```

```
=====================================
```

```
fprintf('MATLAB Simulation Execution Details\n');
```

```
MATLAB Simulation Execution Details
```

```
fprintf('Date and Time: %s\n', char(currentTime));
```

```
Date and Time: 20-10-2026 09:48:52
```

```
fprintf('=====================================\n');
```

```
=====================================
```